

## 9.9 Functions of Matrices, the Exponential and Linear Differential Equations

We have known since Chapter 3 what  $p(\mathbf{A})$  means for a polynomial  $p$ . This section gives a meaning to  $f(\mathbf{A})$  for a much larger class of functions — square roots, logarithms, sines, and above all  $e^{\mathbf{A}}$  — and the Jordan form is what makes it possible. The definition needs a well-definedness proof, which turns out to be Hermite interpolation from Chapter 5: whatever  $f$  is,  $f(\mathbf{A})$  is really  $p(\mathbf{A})$  for a polynomial  $p$  that agrees with  $f$  to the right order at each eigenvalue. With  $e^{t\mathbf{A}}$  in hand, every linear system of differential equations with constant coefficients is solved in one formula.

Throughout this section  $F = \mathbb{C}$  (or  $\mathbb{R}$ , when we say so), so that every characteristic polynomial splits (Corollary 5.5.5) and Jordan forms exist.

### 9.9.1 What a polynomial does to a Jordan block

Start with what we already have. A Jordan block is a scalar plus a nilpotent matrix whose powers stop, and a polynomial applied to it can be read off from a Taylor expansion.

#### Lemma 9.9.1. A Polynomial of a Jordan Block

Let  $F$  have characteristic 0, let  $k \geq 1$ ,  $\lambda \in F$  and  $p \in F[x]$ . Then  $p(\mathbf{J}_k(\lambda))$  is the upper triangular matrix whose  $(r, r+j)$ -entry is

$$(p(\mathbf{J}_k(\lambda)))_{r,r+j} = \frac{p^{(j)}(\lambda)}{j!} \quad (0 \leq j \leq k-1, 1 \leq r \leq k-j),$$

with all entries below the diagonal equal to 0. In particular the value  $p(\lambda)$  fills the diagonal,  $p'(\lambda)$  the superdiagonal, and only the first  $k$  derivatives of  $p$  at  $\lambda$  are involved.

*Proof.* Write  $\mathbf{J} = \mathbf{J}_k(\lambda) = \lambda \mathbf{I}_k + \mathbf{M}$  with  $\mathbf{M} = \mathbf{J}_k(0)$  (Definition 9.3.1). Reading off the columns of  $\mathbf{M}$ , we have  $\mathbf{M}\mathbf{e}_1 = \mathbf{0}$  and  $\mathbf{M}\mathbf{e}_r = \mathbf{e}_{r-1}$  for  $r \geq 2$ , so by induction  $\mathbf{M}^j\mathbf{e}_r = \mathbf{e}_{r-j}$  when  $r > j$  and  $\mathbf{M}^j\mathbf{e}_r = \mathbf{0}$  when  $r \leq j$ . Hence, for  $0 \leq j \leq k-1$ , the matrix  $\mathbf{M}^j$  has 1 in each position  $(r, r+j)$  and 0 elsewhere, and  $\mathbf{M}^k = \mathbf{0}$ . Let  $d = \deg p$ . By Theorem 5.4.7 (b), applied with  $c = \lambda$  and  $n = d$ ,

$$p(x) = \sum_{j=0}^d \frac{p^{(j)}(\lambda)}{j!} (x - \lambda)^j.$$

Substituting  $\mathbf{J}$  for  $x$  — legitimate because evaluation at  $\mathbf{J}$  is a ring homomorphism (Theorem 5.7.1) — and using  $\mathbf{J} - \lambda \mathbf{I}_k = \mathbf{M}$ ,

$$p(\mathbf{J}) = \sum_{j=0}^d \frac{p^{(j)}(\lambda)}{j!} \mathbf{M}^j = \sum_{j=0}^{\min(d, k-1)} \frac{p^{(j)}(\lambda)}{j!} \mathbf{M}^j,$$

since  $\mathbf{M}^j = \mathbf{0}$  for  $j \geq k$ . Each  $\mathbf{M}^j$  contributes its coefficient to every position  $(r, r+j)$  and to no other position, which is the claim. ■

For  $p = x^m$  this reads  $p^{(j)}(\lambda)/j! = \binom{m}{j}\lambda^{m-j}$ , so

$$(\mathbf{J}_k(\lambda)^m)_{r, r+j} = \binom{m}{j}\lambda^{m-j},$$

which is the formula behind everything that follows, here and in the next section.

## 9.9.2 Functions of a matrix

Reading [Lemma 9.9.1](#) backwards suggests the definition. The formula for  $p(\mathbf{J}_k(\lambda))$  uses nothing about  $p$  except the numbers  $p(\lambda), p'(\lambda), \dots, p^{(k-1)}(\lambda)$ . Any function that has those derivatives can be inserted instead.

*To apply  $f$  to a matrix, apply it to each Jordan block, putting  $f^{(j)}(\lambda)/j!$  on the  $j$ -th superdiagonal.*

### Definition 9.9.1. Function of a Matrix

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with Jordan form  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \dots \oplus \mathbf{J}_{k_m}(\lambda_m)$  and  $\mathbf{A} = \mathbf{PJP}^{-1}$ . Let  $f$  be a complex-valued function that is defined and **at least  $k_i - 1$  times differentiable at  $\lambda_i$**  for every  $i$  — for instance, a function given by a convergent power series on a disc around each eigenvalue. Define

$$f(\mathbf{J}_k(\lambda)) := \begin{pmatrix} f(\lambda) & f'(\lambda) & \frac{f''(\lambda)}{2!} & \dots & \frac{f^{(k-1)}(\lambda)}{(k-1)!} \\ & f(\lambda) & f'(\lambda) & \ddots & \vdots \\ & & \ddots & \ddots & \frac{f''(\lambda)}{2!} \\ & & & f(\lambda) & f'(\lambda) \\ & & & & f(\lambda) \end{pmatrix} \in M_k(\mathbb{C}),$$

the upper triangular matrix with  $f^{(j)}(\lambda)/j!$  in every position  $(r, r+j)$ , and

$$f(\mathbf{A}) := \mathbf{P} \left( f(\mathbf{J}_{k_1}(\lambda_1)) \oplus \dots \oplus f(\mathbf{J}_{k_m}(\lambda_m)) \right) \mathbf{P}^{-1}.$$

In words: the diagonal of  $f(\mathbf{J})$  records the values  $f(\lambda_i)$ , and each further superdiagonal records the next derivative, divided by a factorial. For a diagonalizable  $\mathbf{A}$  every block has size 1, no derivatives are needed, and the definition says the familiar thing:  $f(\mathbf{A}) = \mathbf{P} \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_n)) \mathbf{P}^{-1}$ .

**Well-definedness.** Two choices were made: the invertible  $\mathbf{P}$  and the order of the blocks. The Jordan form itself is unique only up to the order of the blocks (Theorem 9.3.1), and even for a fixed  $\mathbf{J}$  there are many  $\mathbf{P}$  with  $\mathbf{A} = \mathbf{PJP}^{-1}$ . The next theorem removes both worries at once, by identifying  $f(\mathbf{A})$  with  $p(\mathbf{A})$  for a suitable **polynomial**  $p$ , an expression in which no choice appears.

### Theorem 9.9.2. Functions of a Matrix Are Polynomials in It

Let  $\mathbf{A} \in M_n(\mathbb{C})$  with distinct eigenvalues  $\mu_1, \dots, \mu_d$ , and write

$$m_{\mathbf{A}} = (x - \mu_1)^{s_1} \dots (x - \mu_d)^{s_d},$$

so that  $s_l$  is the largest size of a  $\mu_l$ -block of the Jordan form of  $\mathbf{A}$  (Corollary 9.3.2 (c)). Let  $f$  be as in [Definition 9.9.1](#). Then:

- (a) there is a polynomial  $p \in \mathbb{C}[x]$  with  $\deg p < \deg m_{\mathbf{A}}$  and

$$p^{(j)}(\mu_l) = f^{(j)}(\mu_l) \quad \text{for all } l = 1, \dots, d \text{ and } j = 0, \dots, s_l - 1,$$

and  $p$  is the **only** such polynomial of degree less than  $\deg m_{\mathbf{A}}$ ;

- (b) for **every** polynomial  $q \in \mathbb{C}[x]$  satisfying those  $\deg m_{\mathbf{A}}$  equations,  $q(\mathbf{A}) = p(\mathbf{A})$ ;  
 (c)  $f(\mathbf{A}) = p(\mathbf{A})$ . In particular  $f(\mathbf{A})$  does not depend on the choice of  $\mathbf{P}$  or on the order of the blocks, and  $f(\mathbf{A})$  is determined by the finitely many numbers  $f^{(j)}(\mu_l)$ .

The polynomial  $p$  of (a) is the **Hermite interpolating polynomial** of  $f$  on the spectrum of  $\mathbf{A}$ .

*Idea.* The list of conditions “match  $f$  and its first  $s_l - 1$  derivatives at  $\mu_l$ ” is exactly a congruence modulo  $(x - \mu_l)^{s_l}$ , by the Taylor expansion; and the moduli  $(x - \mu_l)^{s_l}$  are pairwise coprime. So the Chinese Remainder Theorem produces  $p$  and shows it is unique below the degree of the product  $m_{\mathbf{A}}$  — this is Example (Values and Slopes by CRT) with more nodes and higher orders. Once  $p$  exists, Lemma 9.9.1 says that  $p$  and  $f$  do the same thing to every Jordan block of  $\mathbf{A}$ , because a  $\mu_l$ -block has size at most  $s_l$  and only derivatives up to order  $s_l - 1$  are consulted.

*Proof.*

- (a) For each  $l$  let  $r_l := \sum_{j=0}^{s_l-1} \frac{f^{(j)}(\mu_l)}{j!} (x - \mu_l)^j \in \mathbb{C}[x]$ . For  $g \in \mathbb{C}[x]$ , the Taylor expansion of Theorem 5.4.7 (b) at  $\mu_l$ , together with the uniqueness of the remainder on division by  $(x - \mu_l)^{s_l}$  (Theorem 5.1.1), gives

$$g \equiv r_l \pmod{(x - \mu_l)^{s_l}} \quad \Longleftrightarrow \quad g^{(j)}(\mu_l) = f^{(j)}(\mu_l) \text{ for } j = 0, \dots, s_l - 1,$$

exactly as in Example (Values and Slopes by CRT) (a) for  $s_l = 2$ : the remainder of  $g$  on division by  $(x - \mu_l)^{s_l}$  is  $\sum_{j < s_l} \frac{g^{(j)}(\mu_l)}{j!} (x - \mu_l)^j$ , and two polynomials of degree less than  $s_l$  are equal precisely when their coefficients in the basis  $(1, x - \mu_l, \dots, (x - \mu_l)^{s_l-1})$  agree (Theorem 5.4.7 (a)).

The moduli  $(x - \mu_1)^{s_1}, \dots, (x - \mu_d)^{s_d}$  are pairwise coprime: for  $l \neq l'$ , any common divisor divides both, and a monic common divisor would be a power of  $x - \mu_l$  and a power of  $x - \mu_{l'}$  with  $\mu_l \neq \mu_{l'}$ , hence 1 (Lemma 8.5.2, Theorem 5.3.5). Their product is  $m_{\mathbf{A}}$ . So Theorem 5.6.4 (c) provides exactly one  $p$  with  $\deg p < \deg m_{\mathbf{A}}$  and  $p \equiv r_l \pmod{(x - \mu_l)^{s_l}}$  for every  $l$ , which by the displayed equivalence is (a).

- (b) Let  $q$  satisfy the same equations. By the equivalence again,  $q \equiv r_l \equiv p \pmod{(x - \mu_l)^{s_l}}$  for every  $l$ , so Theorem 5.6.4 (b) gives  $m_{\mathbf{A}} \mid q - p$ . Writing  $q - p = m_{\mathbf{A}} h$  and evaluating at  $\mathbf{A}$ ,  $q(\mathbf{A}) - p(\mathbf{A}) = m_{\mathbf{A}}(\mathbf{A})h(\mathbf{A}) = 0$  since  $m_{\mathbf{A}}(\mathbf{A}) = 0$  (Definition 8.5.1). Hence  $q(\mathbf{A}) = p(\mathbf{A})$ .  
 (c) Let  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \dots \oplus \mathbf{J}_{k_m}(\lambda_m)$  be any Jordan form of  $\mathbf{A}$  and  $\mathbf{A} = \mathbf{PJP}^{-1}$ . Fix  $i$  and let  $l$  be the index with  $\lambda_i = \mu_l$ . By Corollary 9.3.2 (c),  $k_i \leq s_l$ , so Lemma 9.9.1 and the equations in (a) give

$$(p(\mathbf{J}_{k_i}(\lambda_i)))_{r,r+j} = \frac{p^{(j)}(\lambda_i)}{j!} = \frac{f^{(j)}(\lambda_i)}{j!} = (f(\mathbf{J}_{k_i}(\lambda_i)))_{r,r+j} \quad (0 \leq j \leq k_i - 1),$$

the derivatives involved having order at most  $k_i - 1 \leq s_l - 1$ . So  $p(\mathbf{J}_{k_i}(\lambda_i)) = f(\mathbf{J}_{k_i}(\lambda_i))$

for every  $i$ . Since evaluating a polynomial blockwise gives the polynomial of the block diagonal matrix (Theorem 7.1.4 (a), (b)),

$$\begin{aligned} p(\mathbf{A}) &= p(\mathbf{PJP}^{-1}) = \mathbf{P} p(\mathbf{J}) \mathbf{P}^{-1} = \mathbf{P} \left( \bigoplus_i p(\mathbf{J}_{k_i}(\lambda_i)) \right) \mathbf{P}^{-1} \\ &= \mathbf{P} \left( \bigoplus_i f(\mathbf{J}_{k_i}(\lambda_i)) \right) \mathbf{P}^{-1} = f(\mathbf{A}), \end{aligned}$$

the second equality by Proposition 3.7.6 (c). The left-hand side mentions neither  $\mathbf{P}$  nor the order of the blocks, so neither does the right-hand side. This proves the theorem. ■

So the definition is consistent with the old one: if  $f$  happens to be a polynomial, then  $f$  itself satisfies the interpolation conditions, and (b) gives  $f(\mathbf{A})$  in the new sense =  $f(\mathbf{A})$  in the old sense.

*Warning.*  $f(\mathbf{A})$  is not  $f$  applied to the entries of  $\mathbf{A}$ . Even for  $f(x) = x^2$  this fails: with  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , squaring each entry gives  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , while  $\mathbf{A}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ . The definition is designed so that  $f(\mathbf{A})$  is a polynomial in  $\mathbf{A}$ , and entrywise application is not.

### Example. Square Roots of Two Matrices

Using Definition 9.9.1 with  $f(x) = \sqrt{x}$ , the branch with  $\sqrt{4} = 2$  and  $\sqrt{9} = 3$ , compute a square root of

- (a)  $\mathbf{A} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ ;  
 (b)  $\mathbf{B} = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix}$ .

*Solution.*

- (a)  $p_{\mathbf{A}} = (x-5)^2 - 16 = x^2 - 10x + 9 = (x-1)(x-9)$ , so the eigenvalues are 1 and 9, distinct, and  $\mathbf{A}$  is diagonalizable. Solving  $(\mathbf{A} - \mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, -1))$  and  $(\mathbf{A} - 9\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, 1))$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ ,  $\det \mathbf{P} = 2$  and  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ , we have  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1, 9)$  and

$$f(\mathbf{A}) = \mathbf{P} \text{diag}(1, 3) \mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 3 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

*Check.*  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = \mathbf{A}.$

- (b)  $\mathbf{B} = \mathbf{J}_2(4)$  is its own Jordan form, with  $\mathbf{P} = \mathbf{I}$ . Here  $f(4) = 2$  and  $f'(x) = \frac{1}{2\sqrt{x}}$ , so

$f'(4) = \frac{1}{4}$ , and [Definition 9.9.1](#) gives

$$f(\mathbf{B}) = \begin{pmatrix} 2 & \frac{1}{4} \\ 0 & 2 \end{pmatrix}.$$

*Check.*  $\begin{pmatrix} 2 & 1/4 \\ 0 & 2 \end{pmatrix}^2 = \begin{pmatrix} 4 & 2 \cdot \frac{1}{4} + \frac{1}{4} \cdot 2 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} = \mathbf{B}.$

*The Hermite polynomial.* For (b),  $m_{\mathbf{B}} = (x-4)^2$ , and the interpolating polynomial of degree  $< 2$  with  $p(4) = 2$ ,  $p'(4) = \frac{1}{4}$  is  $p = 2 + \frac{1}{4}(x-4) = \frac{1}{4}x + 1$ . Indeed  $p(\mathbf{B}) = \frac{1}{4}\mathbf{B} + \mathbf{I} = \begin{pmatrix} 2 & 1/4 \\ 0 & 2 \end{pmatrix}$ , as [Theorem 9.9.2](#) (c) predicts.

*Quick check.* A matrix  $\mathbf{A} \in M_5(\mathbb{C})$  has Jordan form  $\mathbf{J}_3(2) \oplus \mathbf{J}_1(2) \oplus \mathbf{J}_1(7)$ . Which values of  $f$  and its derivatives does  $f(\mathbf{A})$  depend on?

*Solution.* On  $f(2), f'(2), f''(2)$  and  $f(7)$ : four numbers. Indeed  $m_{\mathbf{A}} = (x-2)^3(x-7)$  by Corollary 9.3.2 (c), so  $s_1 = 3$  for  $\mu_1 = 2$  and  $s_2 = 1$  for  $\mu_2 = 7$ , and [Theorem 9.9.2](#) lists the derivatives of order 0, 1, 2 at 2 and order 0 at 7. Note  $\deg m_{\mathbf{A}} = 4$ , matching the number of conditions and the degree bound on the interpolating polynomial. Nothing about  $f$  away from  $\{2, 7\}$  matters, and no third derivative is consulted even though  $\dim = 5$ .

### 9.9.3 The matrix exponential

The most important function of a matrix is the exponential, and it has a second description, as the sum of a series. We check that the series converges, and that its sum is the matrix  $\exp(\mathbf{A})$  that [Definition 9.9.1](#) produces. Convergence is meant **entrywise**, as in Chapter 8: a sequence  $\mathbf{M}_1, \mathbf{M}_2, \dots$  in  $M_n(\mathbb{C})$  converges to  $\mathbf{M}$  if for each position  $(r, s)$  the number sequence  $(\mathbf{M}_m)_{rs}$  converges to  $\mathbf{M}_{rs}$ , and a series of matrices converges if its sequence of partial sums does. No notion of length of a matrix is used anywhere in this section.

*Remark. What we assume from analysis.* Everything below uses exactly the following facts about real and complex numbers, and nothing else. Chapter 15 will develop the tools that make such statements routine; here they are quoted.

- (A1) **The scalar exponential.** For every  $z \in \mathbb{C}$  the series  $\sum_{i \geq 0} z^i / i!$  converges; its sum is written  $e^z$ . It satisfies  $e^{z+w} = e^z e^w$ ,  $e^0 = 1$ , and  $e^{i\theta} = \cos \theta + i \sin \theta$  for real  $\theta$ . For real  $s \geq 0$ ,  $e^s \geq 1$ .
- (A2) **Termwise differentiation.** If a power series  $\sum_{m \geq 0} c_m t^m$  with  $c_m \in \mathbb{C}$  converges for **every** real  $t$ , then the function  $t \mapsto \sum_m c_m t^m$  is differentiable on  $\mathbb{R}$  with derivative  $\sum_{m \geq 1} m c_m t^{m-1}$ .
- (A3) **Limits and products.** Limits of convergent sequences of complex numbers respect sums, scalar multiples and products. The derivative of a product of two differentiable functions  $\mathbb{R} \rightarrow \mathbb{C}$  obeys the product rule, and differentiation is linear.
- (A4) **Zero derivative.** A differentiable function  $\mathbb{R} \rightarrow \mathbb{C}$  whose derivative vanishes every-

where is constant.

(A5) **Decay.** For every integer  $j \geq 0$  and every real  $c < 0$ ,  $t^j e^{ct} \rightarrow 0$  as  $t \rightarrow \infty$ .

Fact (A5) is used only in the last exercise of this section. Nothing else about  $\exp$ ,  $\cos$  or  $\sin$  is assumed: the derivative  $\frac{d}{dt} e^{zt} = z e^{zt}$  for  $z \in \mathbb{C}$  is the  $1 \times 1$  case of [Theorem 9.9.4](#) (b) below, proved from (A2), and the derivatives of  $\cos$  and  $\sin$  used in the examples follow from it through Euler's formula in (A1).

### Definition 9.9.2. Matrix Exponential

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . The **matrix exponential** of  $\mathbf{A}$  is

$$e^{\mathbf{A}} := \sum_{m=0}^{\infty} \frac{\mathbf{A}^m}{m!} = \lim_{M \rightarrow \infty} \sum_{m=0}^M \frac{\mathbf{A}^m}{m!},$$

the limit taken entrywise, provided it exists;  $\mathbf{A}^0 = \mathbf{I}$ .

That the limit does exist, and what it is, is the content of the next theorem. The Jordan form reduces the question to a single block, where the series can be summed exactly.

### Theorem 9.9.3. The Exponential Series Converges to the Jordan Definition

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Then the series  $\sum_m \mathbf{A}^m/m!$  converges entrywise, and its sum is the matrix  $\exp(\mathbf{A})$  given by [Definition 9.9.1](#) for the function  $f = \exp$ . In particular, for a single Jordan block,

$$e^{\mathbf{J}_k(\lambda)} = e^{\lambda} \begin{pmatrix} 1 & 1 & \frac{1}{2!} & \cdots & \frac{1}{(k-1)!} \\ & 1 & 1 & \ddots & \vdots \\ & & \ddots & \ddots & \frac{1}{2!} \\ & & & 1 & 1 \\ & & & & 1 \end{pmatrix},$$

the upper triangular matrix with  $e^{\lambda}/j!$  in every position  $(r, r+j)$ .

*Idea.* For one block the entries of the partial sums are themselves partial sums of a scalar series, and that series can be recognized: the terms  $\binom{m}{j} \lambda^{m-j}/m!$  simplify to  $\lambda^{m-j}/(j!(m-j)!)$ , which is  $1/j!$  times the  $(m-j)$ -th term of the series for  $e^{\lambda}$ . For general  $\mathbf{A}$ , conjugation by a fixed  $\mathbf{P}$  turns each entry of a partial sum into a fixed finite linear combination of entries of the block partial sums, and limits pass through finite linear combinations.

*Proof. Step 1. One block.* Fix  $k \geq 1$ ,  $\lambda \in \mathbb{C}$  and a position  $(r, r+j)$  with  $0 \leq j \leq k-1$ . By [Lemma 9.9.1](#) applied to  $p = x^m$ ,

$$\left( \frac{\mathbf{J}_k(\lambda)^m}{m!} \right)_{r, r+j} = \frac{1}{m!} \binom{m}{j} \lambda^{m-j} = \begin{cases} \frac{\lambda^{m-j}}{j!(m-j)!}, & m \geq j, \\ 0, & m < j, \end{cases}$$

using  $\binom{m}{j}/m! = 1/(j!(m-j)!)$  for  $m \geq j$  and  $\binom{m}{j} = 0$  for  $m < j$ . Hence the partial sums satisfy

$$\sum_{m=0}^{\mathbf{M}} \left( \frac{\mathbf{J}_k(\lambda)^m}{m!} \right)_{r, r+j} = \frac{1}{j!} \sum_{i=0}^{\mathbf{M}-j} \frac{\lambda^i}{i!} \quad (\mathbf{M} \geq j),$$

after the substitution  $i = m - j$ . By (A1) the right-hand side converges to  $e^\lambda/j!$  as  $\mathbf{M} \rightarrow \infty$ . Entries with  $j < 0$ , that is entries below the diagonal, are 0 in every term. So the series converges entrywise and

$$e^{\mathbf{J}_k(\lambda)} = \left( \text{upper triangular with } \frac{e^\lambda}{j!} \text{ on the } j\text{-th superdiagonal} \right) = \exp(\mathbf{J}_k(\lambda)),$$

the last equality being [Definition 9.9.1](#) for  $f = \exp$ , whose derivatives are  $f^{(j)} = \exp$ , so  $f^{(j)}(\lambda)/j! = e^\lambda/j!$ .

**Step 2. A Jordan matrix.** If  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)$ , then  $\mathbf{J}^m/m!$  is the direct sum of the corresponding blocks (Theorem 7.1.4 (b)), so each entry of a partial sum of  $\sum_m \mathbf{J}^m/m!$  is either identically 0 or an entry of a partial sum from one block. By Step 1 all of them converge, and  $e^{\mathbf{J}} = \bigoplus_i e^{\mathbf{J}_{k_i}(\lambda_i)}$ .

**Step 3. General A.** By Theorem 9.3.1 there are a Jordan matrix  $\mathbf{J}$  and an invertible  $\mathbf{P}$  with  $\mathbf{A} = \mathbf{PJP}^{-1}$ . Then  $\mathbf{A}^m = \mathbf{PJ}^m\mathbf{P}^{-1}$  (Proposition 3.7.6 (c)), so

$$\sum_{m=0}^{\mathbf{M}} \frac{\mathbf{A}^m}{m!} = \mathbf{P} \left( \sum_{m=0}^{\mathbf{M}} \frac{\mathbf{J}^m}{m!} \right) \mathbf{P}^{-1}.$$

Each entry of the left-hand side is a fixed finite linear combination — with coefficients built from the entries of  $\mathbf{P}$  and  $\mathbf{P}^{-1}$ , independent of  $\mathbf{M}$  — of the entries of the middle partial sum. By Step 2 those entries converge, so by (A3) the left-hand side converges entrywise, with

$$e^{\mathbf{A}} = \mathbf{P} e^{\mathbf{J}} \mathbf{P}^{-1} = \mathbf{P} \left( \bigoplus_i e^{\mathbf{J}_{k_i}(\lambda_i)} \right) \mathbf{P}^{-1} = \exp(\mathbf{A}),$$

the last equality by [Definition 9.9.1](#). This proves the theorem. ■

From now on we write  $e^{\mathbf{A}}$  for this matrix and use both descriptions freely. Note one consequence of [Theorem 9.9.2](#):  $e^{\mathbf{A}}$  is a **polynomial** in  $\mathbf{A}$ , namely  $p(\mathbf{A})$  for the Hermite interpolant of  $\exp$  on  $\text{spec}(\mathbf{A})$ . Everything that commutes with  $\mathbf{A}$  therefore commutes with  $e^{\mathbf{A}}$ .

#### Theorem 9.9.4. Properties of the Matrix Exponential

Let  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$  and let  $\mathbf{P} \in M_n(\mathbb{C})$  be invertible.

- (a)  $e^0 = \mathbf{I}$  and  $e^{\mathbf{P}^{-1}\mathbf{A}\mathbf{P}} = \mathbf{P}^{-1}e^{\mathbf{A}}\mathbf{P}$ .
- (b) The function  $\mathbb{R} \rightarrow M_n(\mathbb{C})$ ,  $t \mapsto e^{t\mathbf{A}}$ , is differentiable entrywise, with

$$\frac{d}{dt} e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{A}.$$

- (c)  $e^{\mathbf{A}}$  is invertible, with  $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$ .
- (d) If  $\mathbf{AB} = \mathbf{BA}$ , then  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$ .

$$(e) \det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}}.$$

*Idea.* Part (b) is termwise differentiation, entry by entry. Everything after it is the same two-line trick: to prove  $\mathbf{X}(t) = \mathbf{Y}(t)$  for two matrix functions, show that  $\mathbf{Z}(t) = e^{-t\mathbf{C}}\mathbf{X}(t)$  has derivative 0 for a suitable  $\mathbf{C}$ , so  $\mathbf{Z}$  is constant and equals  $\mathbf{Z}(0)$ . That is why (c) and (d) come after (b), even though the series would also give them. Part (e) is read off the Jordan form:  $e^{\mathbf{J}}$  is triangular with  $e^{\lambda_i}$  on the diagonal.

*Proof.*

- (a)  $0^m = 0$  for  $m \geq 1$  and  $0^0 = \mathbf{I}$ , so the series is  $\mathbf{I}$ . For the second claim, Step 3 of the previous proof applies verbatim with  $\mathbf{J}$  replaced by  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ : the partial sums of  $\sum_m (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^m/m!$  are  $\mathbf{P}^{-1}(\sum_m \mathbf{A}^m/m!)\mathbf{P}$ , and (A3) lets the limit pass through.
- (b) Fix a position  $(r, s)$ . By Theorem 9.9.3 the series  $\sum_m t^m (\mathbf{A}^m)_{rs}/m!$  converges for every real  $t$ , since  $(t\mathbf{A})^m = t^m \mathbf{A}^m$ . So (A2) applies with  $c_m = (\mathbf{A}^m)_{rs}/m!$  and gives

$$\begin{aligned} \frac{d}{dt} (e^{t\mathbf{A}})_{rs} &= \sum_{m \geq 1} \frac{m t^{m-1}}{m!} (\mathbf{A}^m)_{rs} = \sum_{m \geq 1} \frac{t^{m-1}}{(m-1)!} (\mathbf{A} \cdot \mathbf{A}^{m-1})_{rs} \\ &= \left( \mathbf{A} \sum_{l \geq 0} \frac{t^l \mathbf{A}^l}{l!} \right)_{rs}, \end{aligned}$$

where the last step moves the constant matrix  $\mathbf{A}$  outside the limit: each entry of  $\mathbf{A}(\sum_{l \leq L} t^l \mathbf{A}^l/l!)$  is a fixed linear combination of entries of the partial sum, so (A3) applies. Hence  $\frac{d}{dt} e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}}$ . The same computation with  $\mathbf{A}^m = \mathbf{A}^{m-1}\mathbf{A}$  gives  $e^{t\mathbf{A}}\mathbf{A}$ .

- (c) Put  $\mathbf{H}(t) = e^{t\mathbf{A}}e^{-t\mathbf{A}}$  for  $t \in \mathbb{R}$ . Each entry of  $\mathbf{H}$  is a finite sum of products of differentiable functions, so by (A3) and (b),

$$\mathbf{H}'(t) = (\mathbf{A}e^{t\mathbf{A}})e^{-t\mathbf{A}} + e^{t\mathbf{A}}(-\mathbf{A}e^{-t\mathbf{A}}) = \mathbf{A}e^{t\mathbf{A}}e^{-t\mathbf{A}} - \mathbf{A}e^{t\mathbf{A}}e^{-t\mathbf{A}} = 0,$$

where the second equality uses  $e^{t\mathbf{A}}\mathbf{A} = \mathbf{A}e^{t\mathbf{A}}$  from (b). By (A4) every entry of  $\mathbf{H}$  is constant, so  $\mathbf{H}(t) = \mathbf{H}(0) = e^0 e^0 = \mathbf{I}$  for all  $t$ . At  $t = 1$ ,  $e^{\mathbf{A}}e^{-\mathbf{A}} = \mathbf{I}$ , and Theorem 2.4.5 finishes it.

- (d) Suppose  $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ . Then  $\mathbf{B}$  commutes with every partial sum of  $\sum_m t^m \mathbf{A}^m/m!$ , so by (A3) it commutes with the limit:  $\mathbf{B}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{B}$  for every  $t$ . Put

$$\mathbf{G}(t) = e^{-t(\mathbf{A}+\mathbf{B})}e^{t\mathbf{A}}e^{t\mathbf{B}}.$$

Differentiating entrywise with the product rule (A3) and (b),

$$\mathbf{G}'(t) = -(\mathbf{A} + \mathbf{B})e^{-t(\mathbf{A}+\mathbf{B})}e^{t\mathbf{A}}e^{t\mathbf{B}} + e^{-t(\mathbf{A}+\mathbf{B})}\mathbf{A}e^{t\mathbf{A}}e^{t\mathbf{B}} + e^{-t(\mathbf{A}+\mathbf{B})}e^{t\mathbf{A}}\mathbf{B}e^{t\mathbf{B}}.$$

In the third term replace  $e^{t\mathbf{A}}\mathbf{B}$  by  $\mathbf{B}e^{t\mathbf{A}}$ ; in the first, use  $(\mathbf{A} + \mathbf{B})e^{-t(\mathbf{A}+\mathbf{B})} = e^{-t(\mathbf{A}+\mathbf{B})}(\mathbf{A} + \mathbf{B})$  from (b). The three terms become

$$e^{-t(\mathbf{A}+\mathbf{B})}(-(\mathbf{A} + \mathbf{B}) + \mathbf{A} + \mathbf{B})e^{t\mathbf{A}}e^{t\mathbf{B}} = 0.$$

By (A4),  $\mathbf{G}$  is constant, so  $\mathbf{G}(t) = \mathbf{G}(0) = \mathbf{I}$ . Multiplying on the left by  $e^{t(\mathbf{A}+\mathbf{B})}$  and using (c),  $e^{t\mathbf{A}}e^{t\mathbf{B}} = e^{t(\mathbf{A}+\mathbf{B})}$ ; put  $t = 1$ .

(e) Write  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \bigoplus_i \mathbf{J}_{k_i}(\lambda_i)$  a Jordan matrix. By (a) and [Theorem 9.9.3](#),

$$\det e^{\mathbf{A}} = \det(\mathbf{P}e^{\mathbf{J}}\mathbf{P}^{-1}) = \det e^{\mathbf{J}}$$

(Corollary 6.5.7), and  $e^{\mathbf{J}} = \bigoplus_i e^{\mathbf{J}_{k_i}(\lambda_i)}$  is upper triangular with  $e^{\lambda_i}$  repeated  $k_i$  times along the diagonal. By Theorem 6.4.6 and (A1),

$$\det e^{\mathbf{J}} = \prod_i (e^{\lambda_i})^{k_i} = e^{\sum_i k_i \lambda_i} = e^{\text{tr } \mathbf{J}} = e^{\text{tr } \mathbf{A}},$$

the last equality because similar matrices have equal traces (Theorem 3.7.5). This proves the theorem. ■

Part (e) says something striking:  $\det e^{\mathbf{A}} \neq 0$  always, which re-proves (c), and if  $\text{tr } \mathbf{A} = 0$  then  $e^{\mathbf{A}} \in \text{SL}_n(\mathbb{C})$ .

The following corollary is the formula actually used in computations. Note that  $t\mathbf{J}_k(\lambda)$  is **not** a Jordan block for  $k \geq 2$  and  $t \neq 1$ , since its superdiagonal entries are  $t$ ; the commuting law does the work instead.

#### Corollary 9.9.5. Exponential of a Jordan Block, with a Parameter

Let  $k \geq 1$ ,  $\lambda \in \mathbb{C}$  and  $t \in \mathbb{R}$ . Then  $e^{t\mathbf{J}_k(\lambda)}$  is the upper triangular matrix with

$$(e^{t\mathbf{J}_k(\lambda)})_{r, r+j} = \frac{t^j e^{\lambda t}}{j!} \quad (0 \leq j \leq k-1)$$

and zeros below the diagonal. Consequently, if  $\mathbf{A} = \mathbf{PJP}^{-1}$  with  $\mathbf{J} = \mathbf{J}_{k_1}(\lambda_1) \oplus \cdots \oplus \mathbf{J}_{k_m}(\lambda_m)$ , then

$$e^{t\mathbf{A}} = \mathbf{P} \left( e^{t\mathbf{J}_{k_1}(\lambda_1)} \oplus \cdots \oplus e^{t\mathbf{J}_{k_m}(\lambda_m)} \right) \mathbf{P}^{-1}.$$

In particular, for a diagonal  $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ ,  $e^{t\mathbf{D}} = \text{diag}(e^{td_1}, \dots, e^{td_n})$ .

*Proof.* Write  $t\mathbf{J}_k(\lambda) = (t\lambda)\mathbf{I}_k + t\mathbf{M}$  with  $\mathbf{M} = \mathbf{J}_k(0)$ . The two summands commute, since  $\mathbf{I}_k$  commutes with everything, so [Theorem 9.9.4](#) (d) gives  $e^{t\mathbf{J}_k(\lambda)} = e^{(t\lambda)\mathbf{I}_k} e^{t\mathbf{M}}$ . For the first factor,  $((t\lambda)\mathbf{I}_k)^m = (t\lambda)^m \mathbf{I}_k$ , so the partial sums are  $(\sum_{m \leq \mathbf{M}} (t\lambda)^m / m!) \mathbf{I}_k$  and  $e^{(t\lambda)\mathbf{I}_k} = e^{\lambda t} \mathbf{I}_k$  by (A1). For the second,  $\mathbf{M}^j = 0$  for  $j \geq k$ , so the series terminates:

$$e^{t\mathbf{M}} = \sum_{j=0}^{k-1} \frac{t^j}{j!} \mathbf{M}^j,$$

and  $\mathbf{M}^j$  has 1 in each position  $(r, r+j)$ , as computed in the proof of [Lemma 9.9.1](#). Multiplying by the scalar  $e^{\lambda t}$  gives the entry formula. The second display follows from  $t\mathbf{A} = \mathbf{P}(t\mathbf{J})\mathbf{P}^{-1}$  with [Theorem 9.9.4](#) (a) and Theorem 7.1.4 (b), exactly as in Step 2 of the proof of [Theorem 9.9.3](#). The diagonal case is the case in which every block has size 1. ■

*Warning.*  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$  fails when  $\mathbf{A}$  and  $\mathbf{B}$  do not commute. Take

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both square to 0, so the series stop after two terms:  $e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $e^{\mathbf{B}} = \mathbf{I} + \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . Then

$$e^{\mathbf{A}}e^{\mathbf{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = e^{\mathbf{B}}e^{\mathbf{A}}.$$

Since  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ , at most one of the two products can equal  $e^{\mathbf{A}+\mathbf{B}}$ , so the identity must fail for one of the pairs  $(\mathbf{A}, \mathbf{B})$ ,  $(\mathbf{B}, \mathbf{A})$ . (In fact it fails for both:  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{I}$ , so  $e^{\mathbf{A}+\mathbf{B}}$  has all four entries non-zero and none of them equal to 2.) The moral: check  $\mathbf{AB} = \mathbf{BA}$  before splitting an exponential.

## 9.9.4 Linear systems of differential equations

A system of  $n$  linear differential equations with constant coefficients,

$$x'_1 = a_{11}x_1 + \cdots + a_{1n}x_n, \quad \dots, \quad x'_n = a_{n1}x_1 + \cdots + a_{nn}x_n,$$

is the single equation  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  for a vector-valued function  $\mathbf{x}: \mathbb{R} \rightarrow \mathbb{C}^n$ , differentiated entrywise. In one dimension the answer is  $x(t) = e^{ta}x_0$ ; the matrix exponential makes the general answer look exactly the same.

### Theorem 9.9.6. Solution of a Linear System

Let  $\mathbf{A} \in M_n(\mathbb{C})$  and  $\mathbf{x}_0 \in \mathbb{C}^n$ . Then the function

$$\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}_0 \quad (t \in \mathbb{R})$$

is differentiable and satisfies  $\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t)$  for all  $t$  and  $\mathbf{x}(0) = \mathbf{x}_0$ ; and it is the **only** differentiable function  $\mathbb{R} \rightarrow \mathbb{C}^n$  with these two properties.

*Idea.* Existence is [Theorem 9.9.4](#) (b) applied to a fixed vector. For uniqueness, take any solution  $\mathbf{y}$  and multiply it by  $e^{-t\mathbf{A}}$ , the one factor that can undo the growth: the product has derivative 0, hence is constant. This is the same move as in the proofs of parts (c) and (d) above.

*Proof. Existence.* Each entry of  $\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}_0$  is a linear combination of entries of  $e^{t\mathbf{A}}$  with constant coefficients, so  $\mathbf{x}$  is differentiable with  $\mathbf{x}'(t) = (\frac{d}{dt}e^{t\mathbf{A}})\mathbf{x}_0$  by the linearity of differentiation (A3). By [Theorem 9.9.4](#) (b),  $\mathbf{x}'(t) = \mathbf{A}e^{t\mathbf{A}}\mathbf{x}_0 = \mathbf{A}\mathbf{x}(t)$ . And  $\mathbf{x}(0) = e^0\mathbf{x}_0 = \mathbf{x}_0$  by [Theorem 9.9.4](#) (a).

*Uniqueness.* Let  $\mathbf{y}: \mathbb{R} \rightarrow \mathbb{C}^n$  be differentiable with  $\mathbf{y}' = \mathbf{A}\mathbf{y}$  and  $\mathbf{y}(0) = \mathbf{x}_0$ . Put  $\mathbf{z}(t) =$

$e^{-t\mathbf{A}}\mathbf{y}(t)$ . By the product rule (A3) and [Theorem 9.9.4](#) (b) applied to  $-\mathbf{A}$ ,

$$\mathbf{z}'(t) = (-\mathbf{A}e^{-t\mathbf{A}})\mathbf{y}(t) + e^{-t\mathbf{A}}\mathbf{y}'(t) = -\mathbf{A}e^{-t\mathbf{A}}\mathbf{y}(t) + e^{-t\mathbf{A}}\mathbf{A}\mathbf{y}(t) = \mathbf{0},$$

where the last step uses  $e^{-t\mathbf{A}}\mathbf{A} = \mathbf{A}e^{-t\mathbf{A}}$  from [Theorem 9.9.4](#) (b). By (A4) each entry of  $\mathbf{z}$  is constant, so  $\mathbf{z}(t) = \mathbf{z}(0) = \mathbf{x}_0$  for all  $t$ . Multiplying by  $e^{t\mathbf{A}}$  and using [Theorem 9.9.4](#) (c),

$$\mathbf{y}(t) = e^{t\mathbf{A}}\mathbf{z}(t) = e^{t\mathbf{A}}\mathbf{x}_0 = \mathbf{x}(t).$$

This proves the theorem. ■

The formula is only as useful as our ability to compute  $e^{t\mathbf{A}}$ , and the Jordan form does that. Two shapes of answer occur, and they are worth seeing side by side.

### Example. Two Two-Dimensional Systems

Solve  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  with  $\mathbf{x}(0) = (1, 0)$  for

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$  (complex eigenvalues);
- (b)  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$  (one eigenvalue, one block of size 2).

*Solution.*

- (a)  $p_{\mathbf{A}} = (x-1)^2+4$ , with roots  $\lambda = 1 \pm 2i$ , distinct, so  $\mathbf{A}$  is diagonalizable over  $\mathbb{C}$  and every block has size 1. Solving  $(\mathbf{A} - (1+2i)\mathbf{I})\mathbf{x} = \mathbf{0}$ : the first row reads  $-2ix_1 - 2x_2 = 0$ , so  $x_2 = -ix_1$  and  $(1, -i)$  is an eigenvector;  $(1, i)$  is one for  $1-2i$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$  we get  $\det \mathbf{P} = 2i$  and  $\mathbf{P}^{-1} = \frac{1}{2i} \begin{pmatrix} i & -1 \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ , so  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(1+2i, 1-2i)$ . By [Corollary 9.9.5](#),

$$\begin{aligned} e^{t\mathbf{A}} &= \mathbf{P} \text{diag}(e^{(1+2i)t}, e^{(1-2i)t}) \mathbf{P}^{-1} \\ &= \frac{1}{2} \begin{pmatrix} e^{(1+2i)t} + e^{(1-2i)t} & i(e^{(1+2i)t} - e^{(1-2i)t}) \\ -i(e^{(1+2i)t} - e^{(1-2i)t}) & e^{(1+2i)t} + e^{(1-2i)t} \end{pmatrix}. \end{aligned}$$

By (A1),  $e^{(1 \pm 2i)t} = e^t(\cos 2t \pm i \sin 2t)$ , so  $e^{(1+2i)t} + e^{(1-2i)t} = 2e^t \cos 2t$  and  $e^{(1+2i)t} - e^{(1-2i)t} = 2ie^t \sin 2t$ . Substituting, and using  $i \cdot 2i = -2$ ,

$$e^{t\mathbf{A}} = e^t \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}, \quad \mathbf{x}(t) = e^{t\mathbf{A}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^t \begin{pmatrix} \cos 2t \\ \sin 2t \end{pmatrix}.$$

The solution spirals outwards: rotation at angular speed 2, growth at rate 1. The entries are real, as they must be for a real  $\mathbf{A}$  and a real  $\mathbf{x}_0$ .

*Check.*  $\mathbf{x}'(t) = e^t(\cos 2t - 2\sin 2t, \sin 2t + 2\cos 2t)$ , and  $\mathbf{A}\mathbf{x}(t) = e^t(\cos 2t - 2\sin 2t, 2\cos 2t + \sin 2t)$ . They agree, and  $\mathbf{x}(0) = (1, 0)$ .

- (b)  $\mathbf{A} = \mathbf{J}_2(2)$  is its own Jordan form. Writing  $\mathbf{A} = 2\mathbf{I} + \mathbf{M}$  with  $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , the two summands commute and  $\mathbf{M}^2 = 0$ , so [Corollary 9.9.5](#) with  $k = 2$  gives

$$e^{t\mathbf{A}} = e^{2t}(\mathbf{I} + t\mathbf{M}) = e^{2t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

So  $\mathbf{x}(t) = e^{2t}(1, 0)$ . A different initial vector shows the new phenomenon:  $\mathbf{x}(0) = (0, 1)$  gives  $\mathbf{x}(t) = e^{2t}(t, 1)$ , whose first entry is  $te^{2t}$ . **A repeated eigenvalue with a block of size  $k$  produces solutions containing  $t^j e^{\lambda t}$  for  $j < k$ .** This is the same  $t^j \lambda^m$  pattern that Chapter 8 met for recurrences with repeated roots, now in continuous time.

A scalar equation of higher order is a system in disguise, and the translation is the companion matrix of Section 5.

### Example. A Second-Order Equation with a Repeated Root

Solve  $y'' - 4y' + 4y = 0$  with  $y(0) = 1$  and  $y'(0) = 3$ .

*Solution. Make it a system.* Put  $\mathbf{u} = (y, y')$ . Then  $\mathbf{u}' = (y', y'') = (y', 4y' - 4y)$ , so  $\mathbf{u}' = \mathbf{M}\mathbf{u}$  with

$$\mathbf{M} = \begin{pmatrix} 0 & 1 \\ -4 & 4 \end{pmatrix}, \quad \mathbf{u}(0) = (1, 3).$$

Note that  $p_{\mathbf{M}} = x(x - 4) + 4 = x^2 - 4x + 4$  is the polynomial of the equation, and that  $\mathbf{M} = \mathbf{C}(p_{\mathbf{M}})^{\top}$  for the companion matrix  $\mathbf{C}(p_{\mathbf{M}}) = \begin{pmatrix} 0 & -4 \\ 1 & 4 \end{pmatrix}$  of Definition 9.5.3; by Corollary 9.7.7 the two are similar, so which of them one uses is immaterial.

*Exponentiate.*  $p_{\mathbf{M}} = (x - 2)^2$ . Put  $\mathbf{N} = \mathbf{M} - 2\mathbf{I} = \begin{pmatrix} -2 & 1 \\ -4 & 2 \end{pmatrix}$ ; then  $\mathbf{N}^2 = \begin{pmatrix} 4 - 4 & -2 + 2 \\ 8 - 8 & -4 + 4 \end{pmatrix} = 0$ , so  $2\mathbf{I}$  and  $\mathbf{N}$  commute and [Theorem 9.9.4](#) (d) gives, as in the previous example,

$$e^{t\mathbf{M}} = e^{2t}(\mathbf{I} + t\mathbf{N}) = e^{2t} \begin{pmatrix} 1 - 2t & t \\ -4t & 1 + 2t \end{pmatrix}.$$

(No basis change was needed:  $\mathbf{M}$  is a scalar plus a nilpotent matrix on the nose.)

*Read off the answer.*

$$\mathbf{u}(t) = e^{t\mathbf{M}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = e^{2t} \begin{pmatrix} (1 - 2t) + 3t \\ -4t + 3(1 + 2t) \end{pmatrix} = e^{2t} \begin{pmatrix} 1 + t \\ 3 + 2t \end{pmatrix},$$

so  $y(t) = (1 + t)e^{2t}$ .

*Check.*  $y' = e^{2t} + 2(1 + t)e^{2t} = (3 + 2t)e^{2t}$ , which is the second entry of  $\mathbf{u}$ , as it should be, and  $y'(0) = 3$ . Then  $y'' = 2e^{2t} + 2(3 + 2t)e^{2t} = (8 + 4t)e^{2t}$ , and

$$y'' - 4y' + 4y = [(8 + 4t) - 4(3 + 2t) + 4(1 + t)]e^{2t} = 0.$$

The repeated root 2 produces the term  $te^{2t}$ , exactly as the size-2 block predicts.

Chapter 15 will return to  $e^{\mathbf{A}}$  with norms available, which give quantitative bounds instead of exact formulas. The next section stays with exact formulas and asks the discrete-time question: what happens to  $\mathbf{A}^m$ ?

## 9.9.5 Exercises

### 9.9.5.1 A. Check your understanding

#### Exercise. A1

- State the formula for  $p(\mathbf{J}_k(\lambda))$  and say where the factorials come from.
- Define  $f(\mathbf{A})$  for  $\mathbf{A} \in M_n(\mathbb{C})$  with a given Jordan form, and list the two choices whose irrelevance has to be checked.
- Which values of  $f$  and its derivatives does  $f(\mathbf{A})$  depend on? How many numbers is that?
- State the five properties of  $e^{\mathbf{A}}$  proved in [Theorem 9.9.4](#), with their hypotheses.
- True or false:  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$  for all  $\mathbf{A}, \mathbf{B} \in M_n(\mathbb{C})$ . Justify your answer.
- True or false:  $e^{\mathbf{A}}$  can be singular. Justify your answer.

*Solution.*

- $p(\mathbf{J}_k(\lambda))$  is upper triangular with  $p^{(j)}(\lambda)/j!$  in every position  $(r, r+j)$  ([Lemma 9.9.1](#)). The factorials come from the Taylor expansion  $p = \sum_j \frac{p^{(j)}(\lambda)}{j!}(x-\lambda)^j$  ([Theorem 5.4.7](#)), whose  $j$ -th term contributes  $\mathbf{J}_k(0)^j$ , the matrix with ones on the  $j$ -th superdiagonal.
- $f(\mathbf{A}) = \mathbf{P}(\oplus_i f(\mathbf{J}_{k_i}(\lambda_i)))\mathbf{P}^{-1}$  with  $f(\mathbf{J}_k(\lambda))$  as in (a) ([Definition 9.9.1](#)). The choices are the matrix  $\mathbf{P}$  and the order in which the blocks are listed; both are irrelevant because  $f(\mathbf{A}) = p(\mathbf{A})$  for the Hermite interpolating polynomial  $p$  ([Theorem 9.9.2](#)).
- On  $f^{(j)}(\mu)$  for each eigenvalue  $\mu$  and each  $j < s_\mu$ , where  $s_\mu$  is the exponent of  $x - \mu$  in  $m_{\mathbf{A}}$ , that is, the largest  $\mu$ -block size. The total number is  $\sum_\mu s_\mu = \deg m_{\mathbf{A}}$  ([Theorem 9.9.2](#)).
- $e^{\mathbf{P}^{-1}\mathbf{A}\mathbf{P}} = \mathbf{P}^{-1}e^{\mathbf{A}}\mathbf{P}$  for invertible  $\mathbf{P}$ ;  $\frac{d}{dt}e^{t\mathbf{A}} = \mathbf{A}e^{t\mathbf{A}} = e^{t\mathbf{A}}\mathbf{A}$ ;  $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$ ;  $e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}$  **provided**  $\mathbf{AB} = \mathbf{BA}$ ; and  $\det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}}$  with no hypothesis.
- False. With  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{A}^\top$ , the matrices  $e^{\mathbf{A}}e^{\mathbf{B}}$  and  $e^{\mathbf{B}}e^{\mathbf{A}}$  differ, so the identity cannot hold for both orders; see the warning after [Theorem 9.9.4](#).
- False.  $\det e^{\mathbf{A}} = e^{\text{tr } \mathbf{A}} \neq 0$  by [Theorem 9.9.4](#) (e) and (A1), so  $e^{\mathbf{A}}$  is always invertible ([Theorem 6.5.2](#)).

### 9.9.5.2 B. Practice

#### Exercise. B1: Computing exponentials

Compute  $e^{t\mathbf{A}}$  for each of the following.

- (a)  $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ .
- (b)  $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ .
- (c)  $\mathbf{A} = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix} = \mathbf{J}_3(3)$ .

*Solution.*

- (a)  $p_{\mathbf{A}} = (x-1)^2 - 9$ , with roots 4 and  $-2$ . Eigenvectors:  $(\mathbf{A} - 4\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, 1))$ ;  $(\mathbf{A} + 2\mathbf{I})\mathbf{x} = \mathbf{0}$  gives  $\text{span}((1, -1))$ . With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  we have  $\det \mathbf{P} = -2$ ,  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  and  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(4, -2)$ , so by [Corollary 9.9.5](#)

$$e^{t\mathbf{A}} = \mathbf{P} \text{diag}(e^{4t}, e^{-2t}) \mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} e^{4t} + e^{-2t} & e^{4t} - e^{-2t} \\ e^{4t} - e^{-2t} & e^{4t} + e^{-2t} \end{pmatrix}.$$

Check at  $t = 0$ : both diagonal entries give 1 and both off-diagonal ones 0, so  $e^0 = \mathbf{I}$ .

- (b)  $\mathbf{A} = \mathbf{J}_2(5)$ , so [Corollary 9.9.5](#) with  $k = 2$  gives

$$e^{t\mathbf{A}} = e^{5t}(\mathbf{I} + t\mathbf{M}) = e^{5t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

- (c) By [Corollary 9.9.5](#) with  $k = 3$  and  $\lambda = 3$ , or directly from  $\mathbf{A} = 3\mathbf{I} + \mathbf{M}$  with  $\mathbf{M} = \mathbf{J}_3(0)$  and  $\mathbf{M}^3 = 0$ , giving  $e^{t\mathbf{A}} = e^{3t}(\mathbf{I} + t\mathbf{M} + \frac{t^2}{2}\mathbf{M}^2)$ :

$$e^{t\mathbf{A}} = e^{3t} \begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

This is [Definition 9.9.1](#) for  $f(x) = e^{tx}$ : the  $j$ -th superdiagonal carries  $f^{(j)}(3)/j! = t^j e^{3t}/j!$ .

### Exercise. B2: Solving systems

Solve each initial value problem, and verify your answer by differentiating.

- (a)  $x' = x + 3y$ ,  $y' = 3x + y$ , with  $x(0) = 2$ ,  $y(0) = 0$ .
- (b)  $\mathbf{x}' = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} \mathbf{x}$  with  $\mathbf{x}(0) = (0, 1)$ .
- (c)  $y''' = 0$  with  $y(0) = 1$ ,  $y'(0) = 0$ ,  $y''(0) = 2$ , by turning it into a system.

*Solution.*

- (a) This is  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  for the matrix of [Exercise \(B1: Computing exponentials\)](#) (a). By [Theorem 9.9.6](#),

$$\mathbf{x}(t) = e^{t\mathbf{A}} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} e^{4t} + e^{-2t} \\ e^{4t} - e^{-2t} \end{pmatrix},$$

that is,  $x = e^{4t} + e^{-2t}$  and  $y = e^{4t} - e^{-2t}$ . *Check.*  $x' = 4e^{4t} - 2e^{-2t}$  and  $x + 3y = e^{4t} + e^{-2t} + 3e^{4t} - 3e^{-2t} = 4e^{4t} - 2e^{-2t}$ ; similarly for  $y'$ . And  $x(0) = 2$ ,  $y(0) = 0$ .

- (b) Write  $\mathbf{A} = 3\mathbf{R}$  with  $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ , whose eigenvalues are  $\pm i$  (Example (A Real Matrix with Non-Real Eigenvalues)), so those of  $\mathbf{A}$  are  $\pm 3i$ , distinct. Repeating the computation of [Example \(Two Two-Dimensional Systems\)](#) (a) with  $1 \pm 2i$  replaced by  $\pm 3i$  — eigenvectors  $(1, -i)$  for  $3i$  and  $(1, i)$  for  $-3i$  — gives

$$e^{t\mathbf{A}} = \begin{pmatrix} \cos 3t & -\sin 3t \\ \sin 3t & \cos 3t \end{pmatrix}, \quad \mathbf{x}(t) = \begin{pmatrix} -\sin 3t \\ \cos 3t \end{pmatrix}.$$

*Check.*  $\mathbf{x}'(t) = (-3\cos 3t, -3\sin 3t)$  and  $\mathbf{A}\mathbf{x}(t) = (-3\cos 3t, -3\sin 3t)$ ; and  $\mathbf{x}(0) = (0, 1)$ . The solution runs around the unit circle: no growth, since the eigenvalues are purely imaginary.

- (c) Put  $\mathbf{u} = (y, y', y'')$ . Then  $\mathbf{u}' = (y', y'', 0) = \mathbf{M}\mathbf{u}$  with  $\mathbf{M} = \mathbf{J}_3(0)$ , and  $\mathbf{u}(0) = (1, 0, 2)$ . By [Corollary 9.9.5](#) with  $\lambda = 0$  and  $k = 3$ ,

$$e^{t\mathbf{M}} = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{u}(t) = e^{t\mathbf{M}} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 + t^2 \\ 2t \\ 2 \end{pmatrix}.$$

So  $y(t) = 1 + t^2$ . *Check.*  $y' = 2t$ ,  $y'' = 2$ ,  $y''' = 0$ , and the three initial values are  $1, 0, 2$ . The eigenvalue 0 with one block of size 3 gives polynomial solutions  $t^j e^{0t} = t^j$  for  $j < 3$ .

### Exercise. B3: A square root and a logarithm

- (a) Let  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . Compute  $f(\mathbf{A})$  for  $f(x) = \sqrt{x}$  (positive branch) and verify that  $f(\mathbf{A})^2 = \mathbf{A}$ .
- (b) Let  $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ . Compute  $f(\mathbf{B})$  for  $f(x) = \log x$  with  $\log 1 = 0$ , and verify that  $e^{f(\mathbf{B})} = \mathbf{B}$ .
- (c) Explain why the same method produces **no** square root of  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .

*Solution.*

- (a)  $p_{\mathbf{A}} = (x - 2)^2 - 1 = (x - 1)(x - 3)$ , eigenvalues 1, 3. Eigenvectors  $(1, -1)$  for 1 and

(1, 1) for 3. With  $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$  and  $\mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ ,

$$f(\mathbf{A}) = \mathbf{P} \operatorname{diag}(1, \sqrt{3}) \mathbf{P}^{-1} = \frac{1}{2} \begin{pmatrix} 1 + \sqrt{3} & -1 + \sqrt{3} \\ -1 + \sqrt{3} & 1 + \sqrt{3} \end{pmatrix}.$$

Squaring, the diagonal entry is  $\frac{1}{4}[(1 + \sqrt{3})^2 + (\sqrt{3} - 1)^2] = \frac{1}{4}(4 + 2\sqrt{3} + 4 - 2\sqrt{3}) = 2$ , and the off-diagonal entry is  $\frac{1}{4} \cdot 2(1 + \sqrt{3})(\sqrt{3} - 1) = \frac{1}{2}(3 - 1) = 1$ . So  $f(\mathbf{A})^2 = \mathbf{A}$ .

- (b) Here  $m_{\mathbf{B}} = (x - 1)^2$ , since  $(\mathbf{B} - \mathbf{I})^2 = 0 \neq \mathbf{B} - \mathbf{I}$ , so by [Theorem 9.9.2](#) the answer is  $p(\mathbf{B})$  for the polynomial of degree  $< 2$  with  $p(1) = f(1) = 0$  and  $p'(1) = f'(1) = 1$ , namely  $p = x - 1$ . Hence

$$f(\mathbf{B}) = \mathbf{B} - \mathbf{I} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Its square is 0, so  $e^{f(\mathbf{B})} = \mathbf{I} + f(\mathbf{B}) = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \mathbf{B}$ . The Jordan route gives the

same thing: with  $\mathbf{Q} = \operatorname{diag}(2, 1)$  we get  $\mathbf{Q}^{-1}\mathbf{B}\mathbf{Q} = \mathbf{J}_2(1)$ , and  $f(\mathbf{J}_2(1)) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , so

$$f(\mathbf{B}) = \mathbf{Q}f(\mathbf{J}_2(1))\mathbf{Q}^{-1} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

- (c) The matrix is  $\mathbf{J}_2(0)$ , with the single eigenvalue 0 and a block of size 2. [Definition 9.9.1](#) needs  $f$  and  $f'$  at 0, and  $f(x) = \sqrt{x}$  is not differentiable at 0 — the entry  $f'(0)$  does not exist. In fact  $\mathbf{J}_2(0)$  has no square root at all: if  $\mathbf{X}^2 = \mathbf{J}_2(0)$ , then  $\mathbf{X}^4 = \mathbf{J}_2(0)^2 = 0$ , so  $\mathbf{X}$  is nilpotent, so  $\mathbf{X}^2 = 0$  by Proposition 9.2.1 (a) with  $n = 2$ , contradicting  $\mathbf{X}^2 = \mathbf{J}_2(0) \neq 0$ .

### 9.9.5.3 C. Going deeper

#### Exercise. C1: How badly the exponential law fails

Let  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$  in  $M_2(\mathbb{R})$ .

- Compute  $e^{\mathbf{A}}$ ,  $e^{\mathbf{B}}$ ,  $e^{\mathbf{A}}e^{\mathbf{B}}$  and  $e^{\mathbf{B}}e^{\mathbf{A}}$ .
- Compute  $e^{\mathbf{A}+\mathbf{B}}$  in closed form. *Hint:*  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{I}$ , so split the series into even and odd terms.
- Verify  $\det e^{\mathbf{A}+\mathbf{B}} = e^{\operatorname{tr}(\mathbf{A}+\mathbf{B})}$  and  $\det(e^{\mathbf{A}}e^{\mathbf{B}}) = e^{\operatorname{tr} \mathbf{A}}e^{\operatorname{tr} \mathbf{B}}$ , and explain why equal determinants do not rescue the identity.
- Show that  $e^{\mathbf{A}}e^{\mathbf{B}} = e^{\mathbf{C}}$  for **some**  $\mathbf{C} \in M_2(\mathbb{R})$ , and find  $\operatorname{tr} \mathbf{C}$ .

*Solution.*

- (a)  $\mathbf{A}^2 = \mathbf{B}^2 = 0$ , so  $e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $e^{\mathbf{B}} = \mathbf{I} + \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ . Then

$$e^{\mathbf{A}}e^{\mathbf{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad e^{\mathbf{B}}e^{\mathbf{A}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

- (b) Let  $\mathbf{K} = \mathbf{A} + \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Then  $\mathbf{K}^2 = \mathbf{I}$ , so  $\mathbf{K}^{2l} = \mathbf{I}$  and  $\mathbf{K}^{2l+1} = \mathbf{K}$ . Splitting the partial sums by parity, which is legitimate because each entry sequence is a sum of two convergent sequences (A3),

$$\begin{aligned} e^{\mathbf{K}} &= \left( \sum_{l \geq 0} \frac{1}{(2l)!} \right) \mathbf{I} + \left( \sum_{l \geq 0} \frac{1}{(2l+1)!} \right) \mathbf{K} \\ &= \frac{e + e^{-1}}{2} \mathbf{I} + \frac{e - e^{-1}}{2} \mathbf{K} = \begin{pmatrix} \frac{e+e^{-1}}{2} & \frac{e-e^{-1}}{2} \\ \frac{e-e^{-1}}{2} & \frac{e+e^{-1}}{2} \end{pmatrix}, \end{aligned}$$

where the two scalar sums were identified by adding and subtracting the series for  $e^1$  and  $e^{-1}$  term by term (A1). Numerically the diagonal entries are about 1.543, not 2, so  $e^{\mathbf{A}+\mathbf{B}} \neq e^{\mathbf{A}}e^{\mathbf{B}}$ .

- (c)  $\text{tr}(\mathbf{A} + \mathbf{B}) = 0$  and  $\det e^{\mathbf{A}+\mathbf{B}} = \left(\frac{e+e^{-1}}{2}\right)^2 - \left(\frac{e-e^{-1}}{2}\right)^2 = \frac{1}{4}[(e+e^{-1})^2 - (e-e^{-1})^2] = \frac{1}{4} \cdot 4 = 1 = e^0$ . Also  $\det(e^{\mathbf{A}}e^{\mathbf{B}}) = 2 - 1 = 1 = e^0e^0$ . The determinants agree because both matrices are products of exponentials of trace-zero matrices; a determinant is one number and cannot distinguish two matrices, so it gives no information about the identity either way. The entries do.

- (d)  $\mathbf{X} = e^{\mathbf{A}}e^{\mathbf{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  is symmetric with  $\det \mathbf{X} = 1$  and  $\text{tr} \mathbf{X} = 3$ , so  $p_{\mathbf{X}} = x^2 - 3x + 1$ , whose roots  $\frac{3 \pm \sqrt{5}}{2}$  are real, distinct and positive. So  $\mathbf{X} = \mathbf{P} \text{diag}(\alpha, \beta) \mathbf{P}^{-1}$  with  $\alpha\beta = 1$ ,  $\alpha, \beta > 0$ . Put  $\mathbf{C} = \mathbf{P} \text{diag}(\log \alpha, \log \beta) \mathbf{P}^{-1}$ , a real matrix because  $\mathbf{P}$  can be taken real (the eigenvalues are real) and the logarithms are real. By Definition 9.9.1 applied to  $f = \exp$ ,  $e^{\mathbf{C}} = \mathbf{P} \text{diag}(\alpha, \beta) \mathbf{P}^{-1} = \mathbf{X}$ . Finally  $\text{tr} \mathbf{C} = \log \alpha + \log \beta = \log(\alpha\beta) = \log 1 = 0$ , as Theorem 9.9.4 (e) requires, since  $\det \mathbf{X} = 1$ .

### Exercise. C2: When does every solution die out?

Let  $\mathbf{A} \in M_n(\mathbb{C})$ . Prove that the following are equivalent:

- (i) every solution of  $\mathbf{x}' = \mathbf{A}\mathbf{x}$  satisfies  $\mathbf{x}(t) \rightarrow \mathbf{0}$  entrywise as  $t \rightarrow \infty$ ;
- (ii) every eigenvalue  $\lambda$  of  $\mathbf{A}$  has  $\text{Re } \lambda < 0$ .

*Hint: compute the entries of  $e^{t\mathbf{J}_k(\lambda)}$  and use (A5). For the failing direction, an eigenvector gives a solution you can write down.*

*Solution.*

- (ii)  $\Rightarrow$  (i). By Theorem 9.9.6 every solution is  $\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}_0$ , so it suffices to show  $e^{t\mathbf{A}} \rightarrow 0$  entrywise. Write  $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$  with  $\mathbf{J} = \bigoplus_i \mathbf{J}_{k_i}(\lambda_i)$  (Theorem 9.3.1). Each entry of  $e^{t\mathbf{A}} = \mathbf{P}e^{t\mathbf{J}}\mathbf{P}^{-1}$  is a fixed linear combination of entries of  $e^{t\mathbf{J}}$ , so by (A3) it is enough that  $e^{t\mathbf{J}} \rightarrow 0$ . By Corollary 9.9.5, the non-zero entries of  $e^{t\mathbf{J}_k(\lambda)}$  are

$$\frac{t^j e^{\lambda t}}{j!} \quad (0 \leq j \leq k-1).$$

Writing  $\lambda = c + i\omega$  with  $c = \text{Re } \lambda < 0$ , we get  $|t^j e^{\lambda t}| = t^j e^{ct} |e^{i\omega t}| = t^j e^{ct}$  for  $t \geq 0$ , since  $|e^{i\omega t}| = |\cos \omega t + i \sin \omega t| = 1$  by (A1). By (A5) this tends to 0, and a complex sequence tends to 0 exactly when its moduli do. Hence  $e^{t\mathbf{J}} \rightarrow 0$ .

- (iii)  $\Rightarrow$  (ii). Suppose some eigenvalue  $\lambda$  has  $\operatorname{Re} \lambda = c \geq 0$ , and let  $\mathbf{v} \neq \mathbf{0}$  be an eigenvector. Then  $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$  is a solution: it is differentiable with  $\mathbf{x}'(t) = \lambda e^{\lambda t} \mathbf{v} = e^{\lambda t} \mathbf{A} \mathbf{v} = \mathbf{A} \mathbf{x}(t)$ , and  $\mathbf{x}(0) = \mathbf{v}$ . Pick a coordinate with  $v_r \neq 0$ ; then

$$|x_r(t)| = |e^{\lambda t}| |v_r| = e^{ct} |v_r| \geq |v_r| > 0$$

for every  $t \geq 0$ , because  $e^{ct} \geq 1$  when  $c \geq 0$ . So  $x_r(t)$  does not tend to 0, and (i) fails. This proves the equivalence.